

RESEARCH ARTICLE

Revisiting Keyframe Selection in Learning-Based Dense Monocular SLAM

JAEMIN CHOI^{ID}, JUHWAN BAE^{ID}, AND YUN GU LEE^{ID}

School of Software, Kwangwoon University, Seoul 01897, Republic of Korea

Corresponding author: Yun Gu Lee (harmony96@gmail.com)

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ABSTRACT Recent learning-based dense monocular SLAM systems, such as VGGT-SLAM and MAST3R-SLAM, leverage powerful feed-forward 3D reconstruction models and achieve state-of-the-art performance even in uncalibrated monocular settings. However, in these systems, keyframe selection is often handled by simple frame-level heuristics based on low-level 2D signals, and the impact of input frame composition on feed-forward reconstruction and overall SLAM performance has not been systematically analyzed. In this paper, we reformulate keyframe selection in learning-based dense monocular SLAM as a problem of configuring the input frame set for feed-forward 3D reconstruction, rather than selecting individual frames. From this perspective, we propose a simple segment-level keyframe sampling strategy that aims to stabilize the composition of input frames. We analyze how this input configuration affects reconstruction stability and camera trajectory estimation. Experiments on public benchmarks demonstrate that the proposed strategy consistently improves camera trajectory RMSE compared to the original VGGT-SLAM and MAST3R-SLAM pipelines. These results highlight the importance of input frame configuration in learning-based SLAM and suggest a new perspective on keyframe selection beyond conventional frame-level decisions.

INDEX TERMS Dense monocular SLAM, feed-forward 3-D reconstruction, keyframe selection, learning-based SLAM, visual SLAM.

I. INTRODUCTION

Autonomous navigation requires accurate estimation of the robot's pose together with reconstruction of the surrounding three-dimensional environment [1], [2], [3]. Simultaneous Localization and Mapping (SLAM) has long been studied as a core research problem across a wide range of application domains, including mobile robotics, autonomous driving, and augmented and virtual reality (AR/VR) systems [3], [4], [5], [6], [7]. Early SLAM research focused on probabilistically consistent estimation of robot poses and maps in real-world environments with significant sensor noise and uncertainty. Representative approaches were based on the Extended Kalman Filter (EKF) [3], [8], [9], [10], [11], as well as mapping techniques using LiDAR and range sensors [12]. With the miniaturization of cameras and their emergence as low-cost sensors capable of providing rich visual information, visual SLAM based on monocular and stereo cameras has become a dominant research direction [10], [13].

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Traditional visual SLAM approaches are based on multi-view geometry and bundle adjustment, jointly estimating camera poses and three-dimensional scene structure [14], [15], [16], [17], [18], [19], [20]. Feature-based methods extract keypoints and descriptors from images and estimate camera poses and 3D landmarks through inter-frame matching [21], [22], [23], [24], [25], [26]. The resulting nonlinear optimization problems are typically formulated as factor graphs and solved via online updates or periodic global re-optimization [2], [27]. In monocular SLAM systems, the introduction of keyframes within this optimization framework has enabled compact map representations and controlled computational complexity, allowing long sequences to be processed in a stable manner. Representative systems select new keyframes using relatively hand-crafted heuristics, such as camera translation, rotation, or the number of trackable features, and perform local or global bundle adjustment over the selected keyframes to balance accuracy and real-time performance [28], [29].

Keyframes play a central role in traditional SLAM pipelines, directly influencing map quality, tracking

robustness, and computational cost [28], [30]. When too few keyframes are selected, insufficient viewpoint changes and baselines are often obtained, which can lead to degraded reconstruction quality or accumulated drift [29]. Conversely, selecting an excessive number of keyframes increases graph size and optimization cost, while redundant observations can degrade numerical conditioning [31], [32]. Due to this trade-off, traditional visual SLAM systems have generally adopted design strategies that select a limited number of keyframes using metrics—such as inter-frame displacement, rotation, shared feature counts, or image overlap—while reducing redundancy and maintaining sufficient scene coverage [28], [30], [33].

In recent years, deep learning has been increasingly incorporated into SLAM systems [34], [35], [36]. Feed-forward networks are used to predict depth maps or point clouds directly from monocular images, and the predicted outputs are subsequently integrated into SLAM pipelines. By leveraging pre-trained 3D reconstruction models, recent methods have demonstrated improved performance, including the ability to estimate dense three-dimensional structure and globally consistent camera trajectories using only uncalibrated monocular RGB inputs [37], [38], [39].

One representative example is MAST3R-SLAM [35], which is designed around a two-view 3D reconstruction prior. This design has been reported to achieve strong performance for dense monocular SLAM under uncalibrated settings. In contrast, VGGT-SLAM is built upon multi-view transformer-based feed-forward reconstruction and has demonstrated state-of-the-art performance for dense monocular SLAM under the same setting [36]. In feed-forward SLAM frameworks, the amount and quality of multi-view geometric information that a network can exploit are strongly influenced by how the input frame set is composed and distributed over time. As a result, in learning-based SLAM, keyframes are often treated not as explicit optimization units or design variables, as in classical geometry-based SLAM, but rather as elements of the input frame set provided to the reconstruction network. That is, keyframe selection is frequently absorbed into the problem of determining which frames are supplied as inputs to the feed-forward 3D reconstruction model, rather than being used to directly control the stability of bundle adjustment.

Under this input-centric treatment of keyframes, pre-trained 3D reconstruction models learned from large-scale datasets play a substantial role in determining the overall performance of the SLAM pipeline. Depending on which frames are provided as inputs to the feed-forward reconstruction model, the characteristics of the resulting 3D priors can vary. These variations can potentially affect camera tracking, map fusion, and global optimization outcomes. In particular, when sufficient viewpoint variation is not secured among the input frames, feed-forward reconstruction may operate under geometrically unfavorable conditions. This can lead to fluctuations in reconstruction quality that propagate through the entire SLAM pipeline. Despite these

implications, existing learning-based SLAM systems often determine input frame composition implicitly based on local two-dimensional signals, such as frame-level disparity or optical flow. The impact of such input selection strategies on feed-forward reconstruction and overall SLAM performance has not yet been systematically analyzed.

This work is motivated by these observations and reformulates the keyframe selection problem in learning-based dense monocular SLAM. Rather than treating keyframe selection as a frame-level decision of whether to promote the current frame, we consider it as the problem of constructing an input frame set over a fixed temporal segment for feed-forward 3D reconstruction. Specifically, the central question is not whether a given frame should be selected as a keyframe, but which frames within a segment should be provided as inputs to the feed-forward reconstruction model. To this end, we partition the input video sequence into fixed-length segments. Then, we consider a segment-level keyframe configuration strategy that regularly samples a required number of frames within each segment. Within each segment, simple scene variation is used to adaptively adjust the sampling interval, reducing redundant inputs while maintaining sufficient viewpoint changes.

The contributions of this paper are summarized as follows:

- We reformulate keyframe selection in learning-based dense monocular SLAM as a segment-level input configuration problem rather than a frame-level decision. This reformulation provides a new perspective on the stability of inputs to feed-forward 3D reconstruction models.
- Using the VGGT-SLAM pipeline as a baseline, we isolate the effect of input keyframe configuration by modifying only the frame selection strategy. We systematically analyze the impact of this change on feed-forward reconstruction and overall SLAM performance.
- Through experiments on multiple public benchmarks, we show that the proposed segment-level keyframe configuration achieves more consistent improvements in camera trajectory estimation compared to conventional disparity-based frame-level selection strategies.

II. RELATED WORK

This section reviews keyframe selection strategies in traditional geometry-based SLAM and recent learning-based SLAM approaches.

A. KEYFRAME SELECTION IN CONVENTIONAL SLAM

In traditional geometry-based SLAM, keyframes are treated as explicit design elements that directly influence system performance and map quality, and are selected by jointly considering geometric viewpoint changes, map redundancy, and system load.

1) PTAM [28]

PTAM is a representative keyframe-based SLAM system that separates tracking and mapping into parallel processes under

a monocular camera setting. A new keyframe is inserted when the following conditions are satisfied: (1) a sufficiently high inlier ratio of reprojected map points is observed in the current frame, (2) a minimum number of frames has elapsed since the last keyframe to avoid excessive insertion, and (3) the camera motion relative to existing keyframes is large enough, compared to the average scene depth, to provide a valid stereo baseline.

2) ORB-SLAM [30]

ORB-SLAM adopts a three-thread architecture consisting of tracking, local mapping, and loop closing, with ORB features used as a common representation. Keyframes are selected based on the number of map points reliably tracked in the current frame, the processing status of the local mapping thread, and the proportion of map points shared with a reference keyframe. Subsequently, redundant keyframes are removed using a covisibility graph to control map size.

3) LSD-SLAM [40]

LSD-SLAM is a semi-dense direct SLAM system that operates without explicit feature extraction. A new keyframe is created when a view change measure exceeds a predefined threshold. This measure is computed as a weighted combination of the rotational and translational components of the relative SE(3) transformation between the current reference keyframe and the latest tracking frame. By normalizing the average inverse depth of each keyframe to one, this criterion provides a measure of viewpoint change that is independent of scene scale.

B. LEARNING-BASED SLAM AND KEYFRAMES

Recent learning-based SLAM systems estimate camera poses and 3D structure by using deep learning-based reconstruction models. In this paradigm, keyframes tend to be treated not as carefully designed entities, as in traditional geometry-based SLAM, but rather as elements of the input frame set provided to learning-based models.

1) MAST3R-SLAM

MASt3R-SLAM employs the MASt3R network to predict per-frame point maps and performs pose estimation and local point map fusion through iterative projection-based matching. Keyframe selection is triggered when the number of valid pixel correspondences with the current keyframe, or the number of pixels uniquely matched to that keyframe, falls below predefined thresholds.

2) VGGT-SLAM

VGGT-SLAM takes multiple images as input and predicts camera parameters, depth, and confidence maps for each frame in a feed-forward manner. The coordinate system of the first image is used as a reference. The predicted outputs are then used to construct submaps, which are aligned into a global map through backend optimization. New keyframes are selected by computing Lucas–Kanade–based optical flow

or disparity values with respect to the previous keyframe. The current frame is promoted to a keyframe when these values exceed predefined thresholds.

III. MOTIVATION AND PROBLEM ANALYSIS

A. MOTIVATION

In geometry-based SLAM, keyframe selection has been regarded as an important factor affecting system performance. These approaches aim to form a relatively balanced distribution of keyframes along the camera trajectory by selecting frames that provide sufficiently different viewpoints and information from previous keyframes. Such a distribution facilitates the acquisition of diverse viewpoints and is known to contribute positively to map quality and the stability of global optimization.

In contrast, in learning-based SLAM, keyframes tend to be used primarily as inputs to learning-based reconstruction models rather than as explicit design elements. Research efforts have often focused more on the subsequent learning-based processing stages than on the keyframe selection process itself. Systematic analyses of keyframe selection strategies and their resulting distributions have remained relatively limited. As reviewed in the related work, recent feed-forward approaches such as MASt3R-SLAM and VGGT-SLAM select keyframes by using them as input anchors or by relying on local signals. Under such designs, the influence of keyframe distribution and selection strategy on network estimation results has been less thoroughly examined. To date, there has been no sufficient quantitative analysis of how keyframe selection affects the actual performance of learning-based SLAM systems.

B. PROBLEM ANALYSIS

1) KEYFRAME DENSITY INSTABILITY

In feed-forward SLAM systems such as VGGT-SLAM, the input to the reconstruction network is not composed of all frames but of a subset of selected keyframes. The density of these input frames, or equivalently the frame interval between consecutive keyframes, plays an important role in reconstruction performance. When the frame density is too high, the parallax between consecutive keyframes becomes small, making it difficult to obtain informative 3D geometry. Conversely, when the density is too low, the overlap between keyframes decreases, which complicates reliable correspondence estimation.

In feed-forward SLAM pipelines, keyframes are selected at the frame level based on whether a local signal, such as the disparity with respect to the previous keyframe, exceeds a predefined threshold. In VGGT-SLAM, keyframes determined in this manner are accumulated to form a fixed-length submap input set, which is then provided to the reconstruction network. Under such a frame-level selection scheme, the temporal spacing and distribution of keyframes can vary significantly depending on scene structure and camera motion, even when a fixed threshold is used. As a

result, the rate at which submaps are filled and the timing of submap updates become irregular.

In particular, in segments with large motion magnitude, abrupt motion changes, or complex scene structure, frame-level decision signals can become unreliable. Under such conditions, keyframes may cluster densely in certain intervals or, conversely, become excessively sparse. This leads to an uneven distribution of input keyframes provided to the reconstruction network. Such behavior arises not only from the limitations of individual frame-level decision signals, but also from inherent geometric ambiguities introduced when three-dimensional scenes are projected onto two-dimensional images. As a result, relying solely on local frame-level cues is often insufficient to fully capture scene geometry and camera motion. Keyframe selection strategies that depend heavily on a single frame-level criterion can therefore induce abrupt variations in input frame distribution, which may adversely affect reconstruction or tracking performance.

When the density and distribution of input keyframes vary substantially across sequences or motion conditions, the amount of exploitable multi-view information becomes unstable over time. This instability limits the ability of feed-forward reconstruction models to consistently realize their full potential. The core issue is not the choice of a specific threshold, but a structural limitation in which frame-level criteria are insufficient for stably controlling the submap input configuration.

2) EMPIRICAL ANALYSIS OF THE DISPARITY THRESHOLD AT THE DATASET LEVEL

This subsection provides a quantitative analysis of keyframe density instability in VGGT-SLAM, focusing not on comparing specific disparity thresholds but on examining how disparity-based frame-level thresholding affects keyframe density and trajectory accuracy. To this end, we evaluate three disparity threshold settings: the default value used in VGGT-SLAM ($\tau_{\text{disp}} = 50$), a lower threshold that generates more keyframes ($\tau_{\text{disp}} = 25$), and a higher threshold that generates fewer keyframes ($\tau_{\text{disp}} = 75$). The impact of keyframe density variation is quantitatively evaluated on the TUM RGB-D and 7-Scenes datasets.

Table 1 summarizes the effect of the disparity threshold τ_{disp} on the keyframe ratio (KR) and RMSE. As the threshold increases, the keyframe ratio decreases substantially, while the RMSE remains largely unchanged. This indicates low sensitivity of overall accuracy to the threshold value. More importantly, even when the same disparity threshold is applied, the resulting keyframe ratio differs significantly across datasets. For instance, with $\tau_{\text{disp}} = 50$, 11.54% of frames are selected as keyframes on the TUM dataset, whereas only 4.6% are selected on 7-Scenes. These results indicate that a single disparity threshold is insufficient to consistently predict or control the resulting keyframe density. In other words, treating keyframe selection solely as a frame-level threshold tuning problem is inadequate for achieving stable control of the input density in VGGT-SLAM.

TABLE 1. Effect of the disparity threshold τ_{disp} on the keyframe ratio (KR) and RMSE of VGGT-SLAM ($w = 8$) on the TUM RGB-D and 7-Scenes datasets.

τ_{disp}	TUM dataset		7-scenes dataset	
	KR (%)	RMSE (m)	KR (%)	RMSE (m)
25	23.34	0.043	10.1	0.060
50	11.54	0.040	4.6	0.067
75	5.56	0.043	2.23	0.069

3) SEQUENCE-LEVEL VARIABILITY OF THE DISPARITY THRESHOLD

This subsection analyzes how the disparity threshold τ_{disp} affects performance across different sequences on the TUM RGB-D dataset. The goal is to examine whether a single disparity threshold exhibits consistent behavior across sequences with varying characteristics.

As shown in Table 2, the average RMSE differences across different values of τ_{disp} are relatively small. However, the threshold that yields the best performance varies across sequences. This indicates that, due to differences in camera motion patterns and scene structure, the same disparity threshold does not necessarily lead to the same performance trend across sequences. These observations suggest that the appropriate value of the disparity threshold can differ from one sequence to another.

IV. PROPOSED METHOD

1) SEGMENT-LEVEL REFORMULATION OF KEYFRAME SELECTION

As discussed in the preceding analysis, making keyframe decisions independently at the frame level has inherent limitations in maintaining stable input keyframe density and distribution when scene structure and camera motion vary over time. To address this limitation, we reformulate keyframe selection as determining an appropriate frame sampling density over a fixed temporal segment, rather than deciding whether individual frames should be selected as keyframes. In other words, the key question is no longer whether a specific frame should be selected as a keyframe. Instead, it is how densely frames should be sampled within a given segment.

This perspective is conceptually analogous to approaches in video coding, where a frame is partitioned into blocks and further subdivided into sub-blocks. Appropriate partitioning or prediction modes are then selected based on local characteristics [41], [42], [43]. In our formulation, multiple consecutive frames are grouped into a temporal segment, and one sampling mode is selected from a set of candidate modes with different frame intervals according to the characteristics of that segment. Low-level signals such as disparity are not used to directly select individual keyframes, but can instead serve as cues for evaluating and selecting an appropriate sampling mode at the segment level.

TABLE 2. Sequence-wise RMSE of absolute trajectory error (ATE) for VGGT-SLAM on the TUM RGB-D dataset under different disparity threshold settings (τ_{disp}), with the submap size fixed to $w = 8$. Sim(3) alignment is used. Lower is better.

τ_{disp}	Sequence									
	360	desk	desk2	floor	plant	room	rpy	teddy	xyz	Avg
25	0.059	0.023	0.038	0.044	0.058	0.076	0.025	0.046	0.016	0.043
50	0.070	0.026	0.030	0.048	0.026	0.085	0.024	0.035	0.015	0.040
75	0.061	0.054	0.042	0.050	0.030	0.063	0.028	0.046	0.014	0.043

By reformulating keyframe selection as a segment-level frame density control problem, the proposed approach provides a general basis for stabilizing input frame configurations in feed-forward SLAM pipelines. Its concrete instantiation within a specific SLAM framework, along with the overall architecture, sampling strategies, and mode decision rules, is described in the following subsection.

2) FRAMEWORK OVERVIEW

This subsection conceptually describes how the proposed segment-level keyframe density control is instantiated within the VGGT-SLAM pipeline. VGGT-SLAM is used as a testbed in this work, as its feed-forward reconstruction explicitly relies on a fixed-length set of input keyframes. The proposed framework does not modify the overall structure of VGGT-SLAM. It replaces the existing disparity-based frame-level keyframe selection module with a segment-level sampling framework.

When N_S consecutive frames are received, they are treated as a single temporal segment. For a given segment, the proposed framework selects one sampling mode from a set of candidate modes with different frame intervals. The selected sampling mode is fixed for the duration of the current segment, and keyframes are sampled at a uniform interval accordingly.

In the proposed approach, keyframe selection is not performed independently at each frame. Instead, it is controlled by a global decision made at the segment level. As a result, rapid fluctuations in keyframe density caused by local variations in scene structure or camera motion are mitigated. Rather, the overall scene characteristics and camera motion observed within a segment determine the keyframe density. The selected keyframes are then assembled into a submap input set of length w and provided to VGGT in the same manner as in the original VGGT-SLAM pipeline.

The proposed framework preserves the core estimation components of VGGT-SLAM while reformulating the keyframe selection stage as a segment-level density control problem. Details of the segment definition, candidate sampling modes, and mode decision rules are described in the following subsections.

3) SEGMENT FORMATION AND SAMPLING MODES

This subsection describes the construction of temporal segments and the definition of candidate sampling modes required to apply the proposed framework in practice.

As frames are received sequentially, every N_S consecutive frames are accumulated and defined as a single temporal segment. Each segment is constructed in a non-overlapping manner using consecutive blocks of N_S frames.

For each temporal segment, the proposed method selects one sampling mode from a set of candidate sampling modes. Here, a sampling mode refers to the frame interval used to select keyframes within a segment. Each mode has a predefined interval value. The selected sampling mode is applied consistently within the current segment and does not vary on a per-frame basis. Under this formulation, a temporal segment serves as the decision unit for sampling mode selection, while the sampling mode determines the keyframe density within that segment.

We illustrate the proposed procedure using a simple numerical example. Assume a segment length of $N_S = 128$, a VGGT submap size of $w = 8$, and two possible sampling modes with intervals of 16 and 8. In VGGT-SLAM, each submap input consists of $w + 1$ frames, including one anchor keyframe and w additional keyframes. In this example, frame index 0 is used as the initial anchor frame.

The first segment consists of frames with indices from 1 to 128. If the sampling mode for this segment is determined to be 16, eight keyframes are selected within the segment as $K_1 = \{16, 32, 48, \dots, 112, 128\}$. Here, K_s denotes the set of keyframes selected from the s -th segment. The VGGT submap input is then constructed using the initial anchor frame 0 together with the keyframes in K_1 . That is, $\{0\} \cup K_1$ is used as the input to VGGT. The last keyframe of this submap, frame 128, becomes the anchor frame for the next submap.

The second segment consists of frames with indices from 129 to 256. If the sampling mode is again determined to be 16, the selected keyframes are $K_2 = \{144, 160, 176, \dots, 240, 256\}$. To maintain continuity in VGGT-SLAM, the VGGT submap input constructed from the second segment includes the last keyframe of the previous submap, frame 128, as the anchor keyframe. Thus, $\{128\} \cup K_2$ is used as the input to VGGT.

Alternatively, if the sampling mode for the second segment is determined to be 8, a higher keyframe density is produced within the segment. In this case, the selected keyframes are $K_2 = \{136, 144, 152, \dots, 248, 256\}$. As a result, two VGGT submaps are constructed sequentially, each including the last keyframe of the previous submap as the anchor keyframe. For example, $S_2^1 = \{128, 136, 144, \dots, 184, 192\}$ and

$S_2^2 = \{192, 200, 208, \dots, 248, 256\}$ are used as VGGT inputs. Here, S_s^i denotes the i -th VGGT submap input constructed from the s -th segment.

In this manner, temporal segments are non-overlapping at the frame level, while continuity is preserved at the VGGT submap level through shared anchor keyframes.

4) SEGMENT-LEVEL SAMPLING DECISION

In this work, the disparity information computed in VGGT-SLAM is reinterpreted as a decision signal for determining segment-level sampling density, rather than directly selecting keyframes at the frame level. The key idea is to define a metric that summarizes motion characteristics within a consecutive frame segment and to determine the sampling stride accordingly.

a: SEGMENT-LEVEL DISPARITY REPRESENTATION

We define the disparity between the i -th frame and the $(i+1)$ -th frame as a two-dimensional displacement vector $d_{i,i+1}$. Within a segment consisting of N_S consecutive frames, the accumulated disparity vectors are defined as

$$D_k = \sum_{i=0}^{k-1} d_{i,i+1}, \quad (k = 1, \dots, N_S - 1). \quad (1)$$

Here, D_k represents the accumulated disparity from the beginning of the segment to the k -th frame. As the temporal gap between frames increases, the overlap between frames decreases, which can make direct disparity estimation unreliable or even infeasible. For this reason, we use the accumulated disparity D_k as an indirect representation of motion.

The total accumulated displacement over an entire segment is defined as

$$D_{\text{seg}} = D_{N_S-1}. \quad (2)$$

D_{seg} reflects the overall motion magnitude within the segment.

b: SAMPLING DECISION RULE

In the proposed method, D_{seg} is computed when a segment ends and is used to determine the sampling stride. When D_{seg} is sufficiently large, increasing the frame interval can rapidly reduce the overlap between input frames provided to VGGT. Based on this observation, we adopt the following simple decision rule:

$$\text{stride} = \begin{cases} a, & \text{if } D_{\text{seg}} > \tau_{\text{seg}}, \\ b, & \text{otherwise.} \end{cases} \quad (3)$$

Once the stride is determined, keyframes within the segment are selected through regular sampling according to the chosen stride. Subsequent submap construction and backend optimization in VGGT-SLAM follow the original pipeline without modification. In this way, the proposed

segment-level sampling decision incorporates motion characteristics observed at the segment level, rather than relying excessively on frame-level thresholds.

The decision rule is deliberately kept simple, as the primary objective of this work is to isolate and validate the structural impact of segment-level input density control rather than to introduce additional algorithmic complexity.

For clarity, the overall segment-level keyframe selection procedure is summarized in Algorithm 1.

Algorithm 1 Segment-Level Keyframe Selection

Input: Image Sequence $\{I_t\}$, segment length N_S , threshold τ_{seg} , strides $\{a, b\}$

Output: Keyframe set $\mathcal{K}$

- 1) Partition the image sequence into non-overlapping segments of length N_S .
 - 2) For each segment:
 - a) Compute the total accumulated displacement D_{seg} using Eqs. (1) and (2).
 - b) If $D_{\text{seg}} > \tau_{\text{seg}}$, set stride = a ; otherwise set stride = b .
 - c) Select keyframes using regular sampling with the chosen stride.
 - 3) Construct submaps and proceed with the original VGGT-SLAM pipeline.
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V. EXPERIMENTS

A. EXPERIMENTAL SETUP

This section evaluates whether the proposed segment-level keyframe sampling strategy leads to performance improvements within the VGGT-SLAM pipeline. The goal of this work is not to propose a new SLAM system, but to analyze how changes in input keyframe configuration affect overall performance, using VGGT-SLAM as a representative strong feed-forward learning-based SLAM framework. To this end, experiments are conducted within the same VGGT-SLAM pipeline, with only the keyframe selection policy replaced. All other components are kept identical, allowing the effect of the proposed segment-level keyframe configuration to be quantitatively isolated and evaluated.

VGGT-SLAM combines VGGT-based feed-forward multi-view reconstruction with backend alignment, and has recently demonstrated competitive performance under uncalibrated monocular settings. MAST3R-SLAM, another recent learning-based SLAM system that reports strong performance under similar settings, is included as a reference method to provide performance context. In addition, results from DROID-SLAM are presented to indicate the relative position of the evaluated methods with respect to conventional SLAM approaches.

1) DATASETS

Experiments are conducted using RGB images from the TUM RGB-D and 7-Scenes datasets. Although both datasets

are originally provided with RGB-D data, our experiments follow an uncalibrated monocular setting and do not use depth information or camera intrinsic parameters. The TUM RGB-D dataset consists of indoor handheld sequences with motion-capture-based ground-truth camera trajectories, and performance is evaluated using the root mean square error (RMSE) of absolute trajectory error (ATE). The 7-Scenes dataset contains RGB-D sequences captured in seven small indoor environments. Following the VGGT-SLAM evaluation protocol, we report ATE RMSE as well as dense reconstruction quality metrics, including accuracy, completion, and Chamfer distance.

2) IMPLEMENTATION DETAILS

All experiments are conducted on a system equipped with an NVIDIA RTX 5070 Ti GPU, an Intel i7-12700 CPU, and 32 GB of RAM. Our implementation is based on the publicly available VGGT-SLAM codebase, with the proposed segment-level keyframe sampling module added. All other components of the pipeline, including submap inference, alignment, optimization, backend settings, and hyperparameters, are kept identical to the original implementation. The pretrained VGGT network weights provided by the authors are used without modification.

3) PARAMETERS

For VGGT-SLAM, we use Sim(3) alignment and a submap size of $w = 8$, which yield the best performance among the configurations evaluated in the original VGGT-SLAM framework. In the VGGT-SLAM baseline, the disparity threshold τ_{disp} is set to 50, following the default setting of the public implementation.

For the proposed method, the segment length is fixed to $N_S = 128$, and the submap size w is kept identical to the baseline. The proposed segment-level sampling strategy introduces a segment-level threshold τ_{seg} to determine the sampling stride. We consider $\{a, b\} = \{8, 16\}$ with $\tau_{\text{seg}} = 650$.

B. POSE ESTIMATION EVALUATION

Table 3 presents the root mean square error (RMSE) of absolute trajectory error (ATE) on the TUM RGB-D dataset under Sim(3) alignment. The proposed method achieves an average RMSE of 0.030, showing an overall improvement over the VGGT-SLAM baseline. For most sequences, the proposed method yields lower errors than VGGT-SLAM, indicating that modifying the input keyframe configuration can lead to consistent performance improvements across diverse motion patterns. Notably, substantial reductions in ATE RMSE are observed in the *360*, *floor*, and *room* sequences, where the error decreases from 0.070 to 0.046, from 0.048 to 0.030, and from 0.085 to 0.057, respectively. Although the improvement is limited in some sequences, the proposed method exhibits more balanced performance overall. Results on the 7-Scenes dataset are summarized in Table 4. The proposed method achieves a lower average

RMSE of 0.059, compared to 0.067 obtained by VGGT-SLAM.

In addition, the variance and confidence intervals reported in Tables 3 and 4, computed across sequences using the Student's t -distribution, indicate that the proposed method exhibits more stable performance across sequences. While the numerical improvements in RMSE are relatively modest, the significant reduction in variance (Var) across all sequences demonstrates that our strategy ensures higher reliability compared to the baseline.

C. ANALYSIS OF KEYFRAME DENSITY AND PREDICTABILITY

Table 5 summarizes the comparison between the disparity-based frame-level keyframe selection policy of VGGT-SLAM and the proposed segment-level keyframe composition strategy, in terms of the Keyframe Ratio (KR) and ATE RMSE. KR denotes the proportion of frames that are actually used as keyframes for feed-forward 3D reconstruction relative to the total number of input frames. KR thus serves as an indirect indicator of input frame density and computational load.

For VGGT-SLAM, the KR varies according to the disparity threshold τ_{disp} . As τ_{disp} decreases, the number of selected keyframes increases, leading to a higher KR. However, even under the same τ_{disp} , the KR values differ significantly between the TUM dataset and the 7-Scenes dataset. This indicates that a frame-level disparity-based policy struggles to regulate the input frame density in a stable and predictable manner.

In contrast, the proposed segment-level keyframe composition strategy yields similar keyframe ratios of 7.84% and 6.68% on TUM and 7-Scenes, respectively, while also achieving lower RMSE than VGGT-SLAM across all three τ_{disp} settings. This result suggests that how keyframes are structured at the segment level plays a more critical role in feed-forward 3D reconstruction and overall SLAM performance than merely the absolute number of keyframes.

The proposed method exhibits a relatively stable keyframe density that is less dependent on dataset characteristics for a given parameter configuration. This makes the input frame density more predictable in advance, which is advantageous for SLAM system design under computational budget or resource constraints. Overall, these findings highlight that, in learning-based SLAM, keyframe selection should be viewed not only as a frame-level decision problem, but as a composition problem involving the density and structure of the input frame set. This behavior is further illustrated in Fig. 1, which visualizes the keyframe selection patterns over time for representative sequences. While VGGT-SLAM, which relies on frame-level keyframe selection, exhibits irregular and dataset-dependent keyframe patterns, the proposed method produces structured and predictable patterns governed by segment-level composition.

Table 6 presents a comparison of the end-to-end average frames per second (FPS) between VGGT-SLAM and the proposed method. Here, FPS is defined as the total number

TABLE 3. Root mean square error (RMSE) of absolute trajectory error (ATE) on TUM RGB-D (unit: m) using Sim(3) alignment. All methods are evaluated under uncalibrated monocular settings. For VGGT-SLAM and our method, the same Sim(3) alignment and submap size $w = 8$ are used for a fair comparison. For our method, the segment-level keyframe configuration is fixed with $\tau_{\text{seg}} = 650$ across all sequences. Green denotes the best and light green the second best results. Lower is better.

Method (Uncalib.)	Sequence									mean $\pm$ 95% CI	Var
	360	desk	desk2	floor	plant	room	rpy	teddy	xyz		
DROID-SLAM	0.202	0.032	0.091	0.064	0.045	0.918	0.056	0.045	0.012	0.163 $\pm$ 0.222	0.083224
MASt3R-SLAM	0.070	0.035	0.055	0.056	0.035	0.118	0.041	0.114	0.020	0.060 $\pm$ 0.027	0.001204
VGGT-SLAM	0.070	0.026	0.030	0.048	0.026	0.085	0.024	0.035	0.015	0.040 $\pm$ 0.018	0.000548
Ours	0.046	0.018	0.026	0.030	0.027	0.057	0.024	0.029	0.012	0.030 $\pm$ 0.011	0.000189

TABLE 4. Root mean square error (RMSE) of absolute trajectory error (ATE) on 7-Scenes (unit: m) using Sim(3) alignment. All methods are evaluated under uncalibrated monocular settings. For VGGT-SLAM and our method, the same Sim(3) alignment and submap size $w = 8$ are used for a fair comparison. For our method, the segment-level keyframe configuration is fixed with $\tau_{\text{seg}} = 650$ across all sequences. Green denotes the best and light green the second best results. Lower is better.

Method (Uncalib.)	Sequence								mean $\pm$ 95% CI	Var
	chess	fire	heads	office	pumpkin	kitchen	stairs			
DROID-SLAM	0.047	0.038	0.034	0.136	0.166	0.080	0.044		0.078 $\pm$ 0.049	0.00279
MASt3R-SLAM	0.063	0.046	0.029	0.103	0.114	0.074	0.032		0.066 $\pm$ 0.031	0.00111
VGGT-SLAM	0.038	0.027	0.019	0.108	0.144	0.053	0.081		0.067 $\pm$ 0.043	0.00212
Ours	0.039	0.030	0.020	0.095	0.152	0.051	0.025		0.059 $\pm$ 0.045	0.00232

TABLE 5. Keyframe ratio (KR) and absolute trajectory error (ATE) RMSE of VGGT-SLAM with different disparity thresholds and the proposed segment-level keyframe configuration on the TUM RGB-D and 7-Scenes datasets. KR denotes the percentage of frames used as keyframes.

Method	TUM		7-Scenes	
	KR (%)	RMSE	KR (%)	RMSE
VGGT-SLAM ($\tau=25$)	23.34	0.0428	10.10	0.0595
VGGT-SLAM ($\tau=50$)	11.54	0.0399	4.60	0.0673
VGGT-SLAM ($\tau=75$)	5.56	0.0430	2.23	0.0686
Ours	7.84	0.0298	6.68	0.0589

TABLE 6. Keyframe ratio (KR) and end-to-end average FPS of VGGT-SLAM with different disparity thresholds and the proposed segment-level keyframe configuration on the TUM RGB-D and 7-Scenes datasets.

Method	TUM		7-Scenes	
	KR (%)	FPS	KR (%)	FPS
VGGT-SLAM ($\tau=25$)	23.34	24.54	10.10	43.01
VGGT-SLAM ($\tau=50$)	11.54	42.03	4.60	62.91
VGGT-SLAM ($\tau=75$)	5.56	64.51	2.23	77.56
Ours	7.84	47.72	6.68	47.09

of input frames divided by the overall runtime of the entire SLAM pipeline. In VGGT-SLAM, the computational cost is primarily dominated by the VGGT-based feed-forward 3D reconstruction applied only to selected keyframes, and the overall runtime therefore tends to scale approximately with the keyframe ratio (KR). As a result, even under the same disparity threshold, VGGT-SLAM exhibits different FPS values across datasets due to variations in KR. In contrast,

TABLE 7. Dense reconstruction evaluation on 7-Scenes (unit: m). Green denotes best and light green second best. Lower is better.

Method (Uncalib.)	Acc.↓	Completeness↓	Chamfer↓
MASt3R-SLAM	0.068	0.045	0.056
VGGT-SLAM	0.053	0.055	0.054
Ours	0.050	0.059	0.055

the proposed method maintains similar KR values across different datasets, resulting in average FPS that also shows little variation between datasets.

D. DENSE RECONSTRUCTION EVALUATION

Table 7 summarizes the dense reconstruction performance on the 7-Scenes dataset. As shown in the table, the proposed method achieves the lowest error in terms of accuracy, showing a modest improvement over VGGT-SLAM. In contrast, MASt3R-SLAM attains the best performance in terms of completion, while the proposed method exhibits slightly lower completion compared to VGGT-SLAM. For Chamfer distance, the proposed method performs at a level comparable to VGGT-SLAM.

While the proposed segment-level keyframe selection strategy leads to consistent improvements in trajectory estimation, its impact on dense reconstruction quality is more limited. These observations suggest that the primary benefit of the proposed method lies in stabilizing camera motion estimation rather than substantially enhancing dense surface reconstruction.

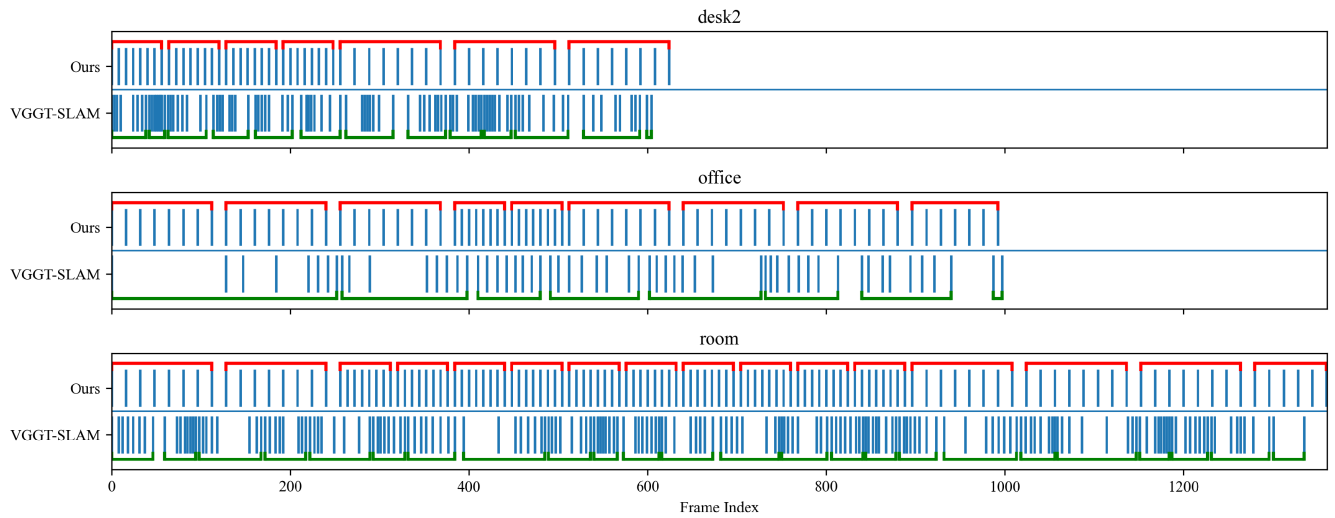


FIGURE 1. Visualization of keyframe selection patterns for VGGT-SLAM and the proposed method on representative sequences from the TUM RGB-D and 7-Scenes datasets. Blue vertical bars indicate selected keyframes along the frame index, while red markers denote submap boundaries ($w = 8$) in the proposed method and green markers indicate submap boundaries used in VGGT-SLAM.

VI. LIMITATIONS

The segment-level regular sampling strategy proposed in this work provides a structured way to control keyframe density at the segment level. However, it also exhibits an implementation-related characteristic in that a predefined number of frames must be accumulated within a segment before the sampling interval and submap processing can be determined. Specifically, in the current implementation, the sampling stride for a segment is decided only after N_S frames have been collected, and submap construction is performed thereafter. Hence, in an online streaming setting, an initial buffering period is required before the first submap can be generated, leading to a latency proportional to the segment length. Rather, it is a common characteristic of many learning-based SLAM systems, including VGGT-SLAM, that rely on submap-level feed-forward inference.

A key distinction of the proposed approach is that the segment length N_S is explicitly defined, making the maximum waiting time predictable in advance. In contrast, frame-level keyframe selection strategies may produce highly variable keyframe densities and submap update timings depending on motion patterns and viewpoint changes. This variability can result in irregular submap update intervals and biased input frame distributions in certain segments.

Another limitation arises when motion characteristics change abruptly within a segment. In the current implementation, the same sampling policy is maintained until the segment boundary is reached, which restricts immediate fine-grained adaptation. This behavior reflects an inherent design trade-off between input stability and strict real-time responsiveness. Extensions such as adaptive segment partitioning, or hybrid strategies that combine frame-level and segment-level decisions could be explored in future work.

In addition, the proposed method restricts the sampling stride to two discrete modes. This is an intentional design choice aimed at balancing input stability and implementation complexity, rather than continuously adjusting the stride or introducing a large number of sampling modes. As demonstrated in the experiments, even with only two stride options, the proposed strategy shows consistent improvement trends in RMSE performance.

For environments with more diverse motion characteristics, expanding the number of stride modes or introducing continuous stride control may offer additional performance gains. However, such extensions would increase both design and computational complexity. Accordingly, this work focuses on demonstrating that meaningful improvements can be achieved with a minimal and simple mode configuration. Future work will explore more diverse motion patterns and adaptive segment lengths to further validate the statistical significance of the proposed strategy in large-scale environments.

VII. CONCLUSION

In this paper, we reformulated keyframe selection in learning-based dense monocular SLAM not as a problem of deciding individual frames, but as a problem of how to construct the input frame set. Since the performance of pre-trained reconstruction models plays a critical role in the overall accuracy and stability of learning-based SLAM pipelines, we focused on the input-level keyframe configuration itself as an important design factor that directly affects SLAM performance.

Unlike conventional frame-level disparity-based keyframe selection, we considered a strategy that constructs input frames at the segment level and structurally controls the sampling density. Experimental results on the TUM RGB-D

and 7-Scenes benchmarks show that even a simple adjustment of sampling density based on segment-level motion characteristics can lead to more stable performance improvements. These results demonstrate, in a quantitative manner, that structural considerations in input frame configuration can improve learning-based SLAM performance without introducing additional learning or complex optimization.

This study suggests that keyframe selection should be revisited not merely as a heuristic design choice, but as a structural component that directly influences the input stability of feed-forward reconstruction models. As future work, promising directions include extending segment partitioning to better support online settings, combining frame-level and segment-level strategies, and exploring more flexible sampling policies.

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JAEMIN CHOI received the B.S. degree from Kwangwoon University, Seoul, Republic of Korea, in 2026. From 2024 to 2026, he was an Undergraduate Research Assistant with the Visual Computing Laboratory, School of Software, Kwangwoon University. His research interests include 3-D computer vision, simultaneous localization and mapping (SLAM), and semantic SLAM.



JUHWAN BAE received the B.S. degree from Kwangwoon University, Seoul, Republic of Korea, in 2026. From 2024 to 2026, he was an Undergraduate Research Assistant with the Visual Computing Laboratory, School of Software, Kwangwoon University. His research interests include computer graphics, Monte Carlo path tracing, real-time denoising, 3-D computer vision, and simultaneous localization and mapping (SLAM).



YUN GU LEE received the B.S., M.S., and Ph.D. degrees in electrical engineering from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea, in 2000, 2002, and 2006, respectively. From 2006 to 2013, he was with the Media Processing Laboratory, DMC R&D Center, Samsung Electronics, Suwon, Republic of Korea, where he was a Principal Engineer. In 2013, he joined the School of Software, Kwangwoon University, Seoul, Republic of Korea, where he is currently a Professor. His research interests include computer vision and deep learning, with a focus on video stabilization, image enhancement, quantization, simultaneous localization and mapping (SLAM), and object tracking.

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